

# Manipulating Singular Values in Muon’s Update

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**Abstract**—Muon replaces the singular values of the momentum gradient with ones. DynMuon raises them to a scheduled power. We ask whether changing this spectrum improves training or just changes the step size. On a 124M GPT trained for 400M FineWeb tokens, tuned Muon and DynMuon reach the same optimum within seed noise, and DynMuon’s bowl is mainly Muon’s bowl lightly shifted in learning rate. Per-layer routing does not beat tuned Muon, but it reduces loss when the global learning rate is too large. Magnitude-matched controls explain why: removing the schedule’s magnitude hurts, randomizing values at the same magnitude barely does, and reversing their order fails. In this setting, exact-SVD Muon gives no meaningful accuracy gain over Newton-Schulz Muon and is about  $1.5\times$  slower. RelMuon, which derives its update scales from the current weight spectrum, is more stable than AdamW but worse and much slower than Muon. The robust signal is magnitude and scale-to-signal ordering, not a special spectrum.

## I. INTRODUCTION

Modern language models are usually trained with AdamW [1], but a growing line of work revisits matrix-aware optimizers for transformer hidden layers. Muon [2], [3] treats the momentum gradient  $M$  as a matrix. If  $M = U\Sigma V^\top$ , Muon applies the polar factor  $UV^\top$ , which is the steepest-descent direction under a spectral-norm constraint [4]. This is simple and effective, but it also makes a strong choice: every nonzero update singular value is set to one.

DynMuon [5] relaxes that choice with  $U\Sigma^{p_t}V^\top$ , annealing  $p_t$  from a gradient-like update toward a more Muon-like and eventually negative-power update. Related work has warned that the exact singular-value shape may be less special than it appears once update magnitude is controlled [6]. Our question is therefore deliberately narrow: when we manipulate the update spectrum, which effects can we actually support from experiments? Our experiments return one positive answer: per-layer routing of the exponent protects against a mistuned learning rate, and a negative one: at a tuned rate, no spectrum we constructed (scheduled, routed, or weight-derived) beats Muon’s flat one.

We contribute (i) a common update view for Muon, DynMuon, routed DynMuon, RelMuon, Polar Muon, Homogeneous Muon and baseline controls; (ii) magnitude-matched tests that separate shape from step size; (iii) ablations over four routing proxies and two weight-spectrum rules. The DynMuon baseline is reproduced exactly from its publicly available implementation.

## II. METHOD

**A common update family.** For each hidden matrix  $W \in \mathbb{R}^{m \times n}$ , the matrix optimizers form the Nesterov momentum  $M_t = G_t + \mu B_t$  with sum-form buffer  $B_t = \mu B_{t-1} + G_t$ , while

AdamW handles embeddings and scalar parameters in every arm. Implementations may rescale the buffer by a positive constant: this leaves the singular vectors of  $M_t$ , and hence all methods below, unchanged (see Appendix A).

All the methods we study apply

$$W_t = (1 - \eta\lambda)W_{t-1} - \eta\sqrt{m/n}D_t, \quad (1)$$

$$D_t = U \operatorname{diag}(s)V^\top,$$

where  $U, V$  come from the thin SVD of  $M_t$ . Table I shows that the methods differ only in the singular values  $s$  assigned to the update.

| Method               | Scale Source                | Update Scales                  |
|----------------------|-----------------------------|--------------------------------|
| SGD Momentum         | momentum                    | raw momentum spectrum          |
| Muon [2]             | fixed                       | flat polar spectrum            |
| HomogeneousMuon [7]  | normalized momentum + fixed | $(\sigma_i(M)/\ M\ _F)^p$      |
| DynMuon [5]          | momentum + time             | scheduled power spectrum       |
| DynMuon-Route        | momentum + layer proxy      | layer-wise power spectrum      |
| RelMuon              | weight spectrum             | RMS-normalized weight spectrum |
| Control (flat polar) | fixed control               | all equal                      |
| Control (random)     | chance spectrum             | random values, norm-matched    |
| Control (inverted)   | reversed spectrum           | largest scale on weakest mode  |
| Control (Kaon [6])   | chaotic polynomial          | iterative spectral map         |

TABLE I

THE OPTIMIZER FAMILY KEEPS THE MOMENTUM SINGULAR DIRECTIONS AND CHANGES ONLY THE UPDATE SINGULAR VALUES. THE CONTROL ROWS MAKE THE SPECTRUM CONTROLS EXPLICIT WITHOUT CROWDING THE MAIN TEXT; APPENDIX A-G GIVES THEIR FORMULAS.

**Why magnitude matters.** For the power family  $D(p) = U\Sigma^pV^\top$ , Lemma 1 gives  $\|D(p)\|_F^2 = \sum_i \sigma_i^{2p}$ . If  $M$  is normalized to  $\|M\|_F = 1$ , then  $\|D(1)\|_F = 1$ , while  $\|D(0)\|_F = \sqrt{k}$  for rank  $k$ . Lowering  $p$  therefore *changes both shape and update size*. Our magnitude-decoupled controls test whether gains come from shape or from this hidden learning-rate change.

**HomogeneousMuon.** We also test a fixed-power SVD variant based on the idea of [7] that says respecting the ordering of the singular values is the important thing. This is experiment is to test what is the change in performance the closer we get to all singular values being 1:

$$M_t = U\Sigma V^\top, \quad D_t(p) = U(\Sigma/\|M_t\|_F)^pV^\top, \quad (2)$$

$$p \in \{0.25, 0.5, 0.75\}.$$

This is another way to make the update singular values closer to each other: it reduces the ratio between the largest and smallest nonzero singular values while keeping the singular-vector directions fixed. The response curves are in Fig. 5.

| Proxy       | Formula                                      | Intended signal         |
|-------------|----------------------------------------------|-------------------------|
| Stable rank | $\ M\ _F^2 / \ M\ _2^2$                      | spectrum spread         |
| SNR         | $\ M\ _F / \ G - M\ _F$                      | momentum consistency    |
| EMA-SNR     | EMA signal/noise ratio                       | smoothed consistency    |
| Alignment   | $ \langle W, M \rangle  / (\ W\ _F \ M\ _F)$ | weight-update agreement |

TABLE II

ROUTING PROXIES. THEY ARE TREATED AS PROBES; THE DATA CHOOSE THE USEFUL SIGNAL RATHER THAN THE METHOD ASSUMING ONE.

**Routed DynMuon.** DynMuon uses one global clock  $p_t$ . Our routed version lets each matrix lean around that clock:

$$z_{t,\ell} = \frac{x_{t,\ell} - \bar{x}_t}{\bar{\sigma}_t}, \quad (3)$$

$$p_{t,\ell} = \text{clip}(p_t + \text{clip}(\beta z_{t,\ell}, -\ell_{\max}, \ell_{\max}), p_{\min}, p_{\max}).$$

The proxy  $x_{t,\ell}$  is experimental: we test stable rank, SNR, EMA SNR, and weight-momentum alignment (Appendix A-F) instead of choosing a default. The z-score and lean cap are included to keep heavy-tailed proxies from turning the router into a saturated controller.

**RelMuon.** RelMuon keeps Muon’s singular directions but takes the scale vector from the current weight matrix. We test  $s_i \propto \log(1 + \sigma_i(W))$  and  $s_i \propto \sigma_i(W)$ , both RMS-normalized. This resembles a spectral trust-ratio, but only by rank: the  $i$ -th weight singular value is paired with the  $i$ -th update direction.

**Attention-only RelMuon.** This ablation applies RelMuon only to attention projections and sends MLP matrices, embeddings, norms, and biases through AdamW. It tests whether most of the spectral-shaping benefit can be bought with fewer SVDs.

### III. EXPERIMENTS

**Training protocol.** We train a 124M-parameter GPT with 12 layers, width 768, sequence length 1024, and tied embeddings for 1526 steps, or 400M FineWeb tokens [8]. All runs use cosine learning-rate decay with 150 warmup steps, weight decay 0.1, and embedding learning rate  $6 \cdot 10^{-3}$ , so the sweeps isolate the matrix learning rate. The main evidence set contains finished Weights&Biases (W&B) runs with this same budget.

**Evidence sets and reporting rule.** We use four experiment sets. First, LR bowls compare AdamW, Muon, exact-SVD Muon, DynMuon, Route-align, and RelMuon-log1p. Second, routing ablations test stable rank, SNR, EMA-SNR, and alignment proxies, plus paired high-LR stress checks and a short anisotropic-noise check. Third, magnitude controls remove DynMuon’s natural Frobenius-norm profile or randomize/invert the singular values at matched magnitude. Fourth, RelMuon ablations compare log1p, RMS, attention-only, weight-decay, and matched-scaling variants.

Each optimizer family contributes its best completed LR point to the main table: three-run means are used where available, and run-to-run noise on identical configurations is 0.001-0.003 final validation loss. All curves shown in Fig. 1 have their best LR bracketed by worse points on both sides. This keeps boundary optima out of the main bowl. A zero-gain routing run matches the released DynMuon baseline (3.6503

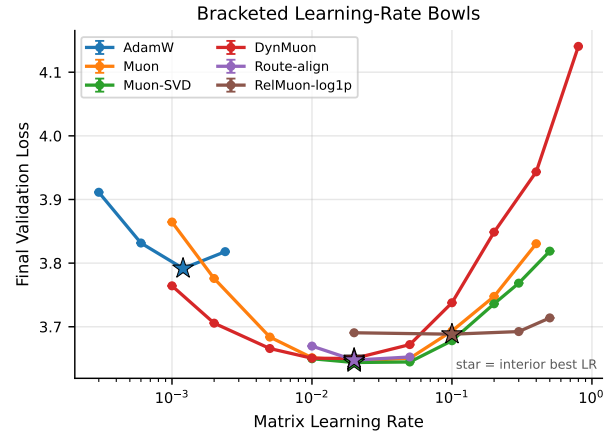


Fig. 1. Bracketed learning-rate bowls (final validation loss vs. matrix learning rate; stars mark each family’s selected point). One figure carries the main comparisons: DynMuon’s curve is Muon’s curve shifted toward smaller rates, exact-SVD Muon sits on top of Newton–Schulz Muon, Route-align hugs the baselines, and RelMuon-log1p plateaus above them.

vs 3.6508), checking implementation parity. We report both final validation loss and seconds per step, because several spectra look acceptable in step count but fail once wall-clock cost is included.

### IV. RESULTS AND DISCUSSION

**The schedule is a learning-rate reparameterization.** Tuned Muon and DynMuon are nearly tied: Muon reaches  $3.6456 \pm 0.0008$  and DynMuon  $3.6508 \pm 0.0008$  at  $\eta = 0.02$  (3 seeds). The informative structure is that their bowls *cross* (Fig. 1): DynMuon is ahead by  $\sim 0.1$  at the smallest tested rate and behind by a similar margin at ten times the optimum. Lemma 1 explains both sides at once: the schedule’s negative-power phase amplifies update norms, so DynMuon behaves like Muon run at a larger learning rate: a free boost when the rate is too small, a liability when it is too large, and a wash when tuned. Fig. 2 shows the same ordering over training: the Muon-family representatives stay within one band, while RelMuon and AdamW separate.

**Flatter spectra help, up to the polar limit.** The fixed-power HomogeneousMuon sweep probes the same conclusion from the constructive side. Since  $\sigma_{\max}^p / \sigma_{\min}^p = (\sigma_{\max} / \sigma_{\min})^p$ , lowering  $p$  homogenizes the update spectrum toward Muon’s flat one (Fig. 5). Final loss improves monotonically as  $p$  decreases, and the strongest homogenization we tested ( $p = 0.25$ ) comes closest to the flat-polar baselines (Appendix Fig. 6). Flattening is the helpful direction; Muon simply takes it to its limit.

**Exact polar flattening is not the bottleneck.** The practical Muon baseline does not apply the exact polar factor: its twelve Newton–Schulz steps act as a soft spectral filter that pushes moderate and large singular values to one but leaves very small modes below one (Appendix Fig. 4), a gap emphasized in recent analyses [9]. Replacing it with an exact-SVD polar update (which is the ideal flattening used as the reference base-

| optimizer representative | LR     | final val                             | s/step |
|--------------------------|--------|---------------------------------------|--------|
| Muon-SVD                 | 0.02   | 3.6438                                | 4.20   |
| Muon (3 seeds)           | 0.02   | <b><math>3.6456 \pm 0.0008</math></b> | 2.79   |
| Route-align              | 0.02   | 3.6476                                | 3.62   |
| DynMuon (3 seeds)        | 0.02   | $3.6508 \pm 0.0008$                   | 3.38   |
| RelMuon-log1p            | 0.1    | 3.6883                                | 10.81  |
| AdamW                    | 0.0012 | 3.7917                                | 2.29   |

TABLE III

SELECTED 1526-STEP REPRESENTATIVES: THE SAME BRACKETED METHODS USED IN FIGS. 1 AND 2. MUON-SVD IS SINGLE-SEED AND USES THE AUXILIARY SVD-VS-NEWTON-SCHULZ PROTOCOL.

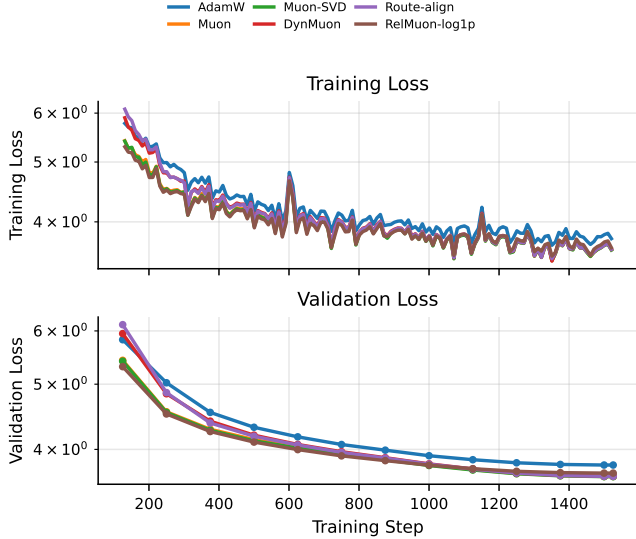


Fig. 2. Training and validation losses for the selected representatives in the LR bowl. Curves start after the shared initialization spike. SVD-polar Muon tracks Muon closely but is slower per step; RelMuon-log1p is competitive early in step count but not in final loss or wall-clock cost.

line in convergence analyses [10]) changes almost nothing: the exact-polar sweep bottoms at 3.6438, within a few seed deviations of the Newton–Schulz mean, at  $1.5\times$  the step cost (Fig. 1). Because this auxiliary sweep also scales the AdamW fallback LR with the matrix LR, we read it as a backend check rather than an optimizer claim; either way, the finite polar approximation is not what limits Muon.

Table III collects the selected representatives; the cost axis changes the interpretation. Muon is not only the best seeded point, it is also the fastest matrix optimizer to the late validation targets, while SVD-polar Muon pays  $1.5\times$  and RelMuon-log1p nearly  $4\times$  per step (Appendix Fig. 13).

**Routing does not improve a tuned optimum, but it protects a mistuned one.** At the tuned rate no proxy opens a new optimum: alignment is the only one that stays inside the baseline noise band, stable-rank routing worsens monotonically with its gain, and the SNR proxy is insensitive to the sign of its gain (Appendix Figs. 7 and 8). The routed exponents are nevertheless structured rather than random (Appendix Fig. 9), so the limitation is the proxy signal, not the routing machinery. The useful case is a mistuned rate: Route-align at  $\eta = 0.05$  recovers  $\sim 0.017$  validation loss over DynMuon, stable-rank

routing recovers  $\sim 0.022$  at  $\eta = 0.2$ , and the same leans lose below the optimum (Appendix Fig. 10). A short anisotropic-noise stress test is neutral. Routing is therefore a step-size safety net, not an accelerator or a noise filter, which also explains why it looks weak at the tuned rate, where the shared schedule leaves layer-local offsets little useful work.

**Magnitude does the work; ordering is the only shape constraint.** The two controls test the same hypothesis from opposite sides. Removing DynMuon’s natural magnitude profile while keeping its routed shape is the most damaging intervention we tried (best grid point 3.7345, Appendix Fig. 10). Keeping the magnitude and *randomizing* the spectrum values barely hurts, but assigning the same values in reverse order fails outright, costing  $+0.32$  (Appendix Fig. 11). The exact values are not sacred; the effective step size and the scale-to-signal ordering are. This qualifies the “Muon is not that special” story [6]: random spectra work only when magnitude is matched and large scales still land on strong momentum directions.

**Weight-derived spectra are a costly conservative bias.** RelMuon-log1p is flat around 3.69 across a  $15\times$  range of learning rates: robust, but  $\sim 0.043$  worse than Muon at nearly  $4\times$  the step cost; adding weight decay or matched shape-LR scaling does not change the direction. RMS RelMuon trails log1p, consistent with raw weight singular values over-emphasizing already-large modes; its sweep is not bracketed, so it is excluded from the main bowl. The attention-only ablation answers its efficiency question negatively: it lands near AdamW-level loss while keeping most of the SVD cost (Appendix Fig. 12). Weight-derived spectra act as conservative regularization, not a competitive Muon replacement.

## V. CONCLUSIONS

On this 400M-token GPT setup, no tested spectrum beats tuned Muon. DynMuon’s schedule mainly reparameterizes the learning rate: helpful below the optimum, harmful above it, and neutral near it. Fixed-power homogenization improves monotonically as the spectrum approaches Muon’s flat one, and exact-SVD flattening adds nothing over the finite Newton-Schulz filter at  $1.5\times$  the cost. Routing gives the learning-rate lesson per layer: it does not improve the tuned optimum, but it recovers 0.017–0.022 validation loss when the global rate is  $2.5\text{--}10\times$  too large. The main methodological point is that spectrum changes must be checked against magnitude-matched controls. Removing DynMuon’s magnitude profile hurts; randomizing values at the same magnitude does not; reversing their order does. RelMuon is stable but slower and worse. *Plain Muon remains the strongest cheap default in this family.*

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## APPENDIX A

### THE UPDATE FAMILY: IDENTITIES BEHIND THE CLAIMS

This appendix backs the Method section and the magnitude claim of §IV: the two lemmas below are, respectively, the reason the exponent doubles as a step size and the reason the optimizer needs no SVD at training time.

#### A. Momentum conventions

For one matrix parameter  $W$  with stochastic gradient  $G_t$ , the DynMuon reference stores a sum-form Nesterov buffer

$$B_t = \mu B_{t-1} + G_t, \quad M_t = G_t + \mu B_t, \quad (4)$$

while other implementations store the EMA-scaled buffer  $\tilde{B}_t = (1 - \mu)B_t$ , giving  $\tilde{M}_t = (1 - \mu)M_t$ . The two conventions differ by a positive constant  $c$ , and the thin SVD of  $cM$  is  $U(c\Sigma)V^\top$ : singular vectors are unchanged, singular values scale by  $c$ . Muon (which discards  $\Sigma$ ), RelMuon (whose scales come from  $W$ , not  $M$ ), and every RMS-normalized variant are therefore convention-invariant; only the raw power family picks up the factor  $c^p$ , which is why our DynMuon path keeps the reference sum convention for exact parity with the released baseline.

A codebase using the EMA-scaled buffer  $\tilde{B}_t = (1 - \mu)B_t$  instead of the sum-form  $B_t$  changes the momentum matrix by a positive constant  $c = (1 - \mu)$ , and by Lemma 1 the update norm scales as  $c^p$ . For  $p \neq 0$ , this is a silent learning-rate change whose magnitude grows with  $|p|$  and with distance from  $c = 1$ . All experiments in this paper use the sum-form convention; practitioners porting any  $p \neq 0$  power variant to a different convention must rescale accordingly. Methods unaffected by this choice are Muon (which discards  $\Sigma$ ), RelMuon (scales come from  $W$ ), and RMS-normalized variants (the constant cancels in the ratio).

#### B. Magnitude is not separable by default

**Lemma 1** (Norm of the shaped update). *For  $D(p) = U\Sigma^p V^\top$ ,*

$$\|D(p)\|_F^2 = \sum_i \sigma_i^{2p}. \quad (5)$$

*Proof.* Write  $D(p) = \sum_{i=1}^k \sigma_i^p u_i v_i^\top$ . For any two indices, the Frobenius inner product of the rank-one terms is

$$\langle u_i v_i^\top, u_j v_j^\top \rangle_F = \text{tr}(v_i u_i^\top u_j v_j^\top) = (u_i^\top u_j)(v_i^\top v_j) = \delta_{ij},$$

since the columns of  $U$  and of  $V$  are orthonormal. The matrices  $\{u_i v_i^\top\}_{i \leq k}$  are therefore an orthonormal system in the Frobenius inner product, and  $\|D(p)\|_F^2 = \sum_i (\sigma_i^p)^2 = \sum_i \sigma_i^{2p}$ .  $\square$

If  $M$  is Frobenius-normalized, then  $p = 1$  gives norm 1 and  $p = 0$  gives  $\sqrt{k}$ . For transformer matrices with  $k$  in the hundreds, this is a large implicit learning-rate change. This is why the magnitude-decoupled control is central: it sets

$$\hat{D}(p) = \sqrt{k} \frac{D(p)}{\|D(p)\|_F} \quad (6)$$

so that spectral shape changes without the natural DynMuon magnitude profile. The exact constant  $\sqrt{k}$  does not matter: changing it only rescales the learning rate, which we sweep anyway. What is truly removed is the variation of  $\|D(p_t)\|_F$  over steps and layers, so the observed degradation measures the value of that variation. Two caveats: the best decoupled point sits at the edge of its LR grid, and the mirror control (flat spectrum at DynMuon’s varying norm) was not run.

**Lemma 2** (Gram-factor identity). *Orient  $M \in \mathbb{R}^{m \times n}$  so that  $m \leq n$ , let  $X = M/\|M\|_F = U\Sigma_n V^\top$  be a thin SVD with  $\Sigma_n = \Sigma/\|M\|_F$ , and let  $A = XX^\top$ . With matrix powers of  $A$  defined by its eigendecomposition (zero eigenvalues mapped to zero),*

$$\|M\|_F^p A^{p/2} UV^\top = U\Sigma^p V^\top. \quad (7)$$

*Proof.* Since  $m \leq n$ ,  $U \in \mathbb{R}^{m \times k}$  with  $k = \min(m, n) = m$  when  $M$  has full row rank; in general complete the columns of  $U$  to an orthonormal basis  $[U \ U_\perp]$  of  $\mathbb{R}^m$ . From  $X = U\Sigma_n V^\top$  and  $V^\top V = I$ ,

$$A = XX^\top = U\Sigma_n^2 U^\top = [U \ U_\perp] \begin{pmatrix} \Sigma_n^2 & 0 \\ 0 & 0 \end{pmatrix} [U \ U_\perp]^\top,$$

which is an eigendecomposition of  $A$ . Applying the spectral map  $\lambda \mapsto \lambda^{p/2}$  (with  $0 \mapsto 0$ ) gives  $A^{p/2} = U\Sigma_n^p U^\top + 0$ . Then, using  $U^\top U = I$ ,

$$A^{p/2} UV^\top = U\Sigma_n^p U^\top U V^\top = U\Sigma_n^p V^\top,$$

and multiplying by  $\|M\|_F^p$  converts the normalized spectrum back to the raw one:  $\|M\|_F^p \Sigma_n^p = \Sigma^p$ . The zero eigenvalues are exactly the directions outside the row space of  $M$ ; mapping them to zero is the pseudo-power convention that keeps  $p \leq 0$  from amplifying null directions.  $\square$

Lemma 2 gives an implementation path through a small Gram eigendecomposition plus a polar factor when  $p \geq 0$ . For  $p < 0$  and rank-deficient  $M$ , the pseudo-power convention (zero eigenvalues mapped to zero) is not sufficient to avoid numerical issues: negative powers of near-zero eigenvalues require a separate regularization path, which the reference DynMuon implementation handles via a Taylor approximation to the Gram factor (described in the following subsection). The unified Gram path therefore applies cleanly only in the  $p \geq 0$  regime.

**Reference DynMuon.** The released DynMuon optimizer uses a three-phase transform: raw momentum for  $p \geq 0.25$ , Newton-Schulz polar updates for  $0 \leq p < 0.25$ , and a Taylor approximation to the negative-power Gram factor for  $p < 0$ . Our reference path preserves those conventions for baseline comparisons, while the SVD path is used for exact spectrum-shape controls. This distinction matters because recent Muon work treats finite polynomial orthogonalization as part of the optimizer, not merely as an implementation detail [10]. The SVD and Gram/Newton-Schulz paths are checked by the validation suite, and Fig. 3 verifies empirically that the two implementations of the same operator agree: a single shaped update matches the exact SVD to  $\sim 10^{-6}$  relative error (30-step cubic iteration) across the whole exponent range, and full

30-step optimizer trajectories computed in the two modes stay within  $10^{-6}$  relative weight divergence.

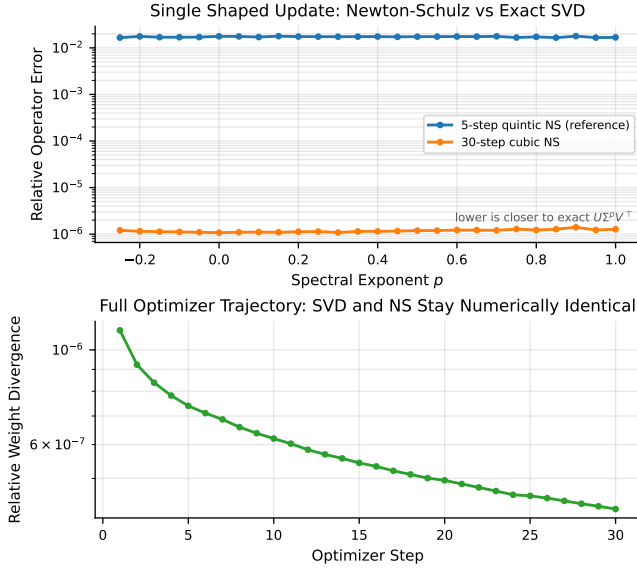


Fig. 3. Empirical SVD  $\equiv$  Newton-Schulz equivalence for  $D(p)$ . Top: relative error of one shaped update vs. the exact SVD across exponents (the reference 5-step quintic iteration is a coarser fixed-cost approximation at  $\sim 2 \cdot 10^{-2}$ ). Bottom: relative divergence between full optimizer trajectories run in the two compute modes.

### C. Finite Newton-Schulz response

The practical Muon baseline first normalizes the momentum matrix by its Frobenius norm and then applies twelve steps of the quintic Newton-Schulz map. On singular values this is a smooth finite-step response, not the discontinuous exact-polar replacement  $x \mapsto 1$ . For very small  $x$ ,  $f(x) \approx \frac{15}{8}x$ , so each step scales tiny modes by approximately 1.875; after twelve steps a mode at  $x = 0.001$  reaches roughly  $1.875^{12} \times 0.001 \approx 0.14$ , well below one (Fig. 4). This is the spectral filter that the exact-SVD comparison of §IV replaces with ideal flattening, and that comparison shows the difference does not matter for final loss.

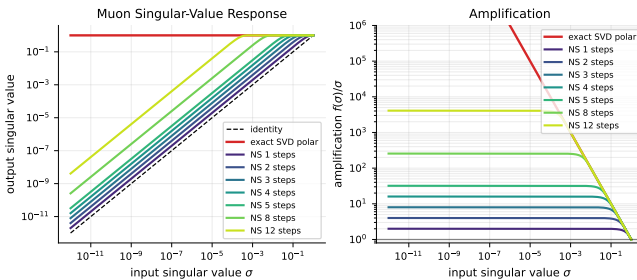


Fig. 4. Singular-value response of the repo's twelve-step quintic Newton-Schulz Muon map. Moderate and large modes approach the polar response, while very small modes remain below one.

### D. Fixed-power HomogeneousMuon response

HomogeneousMuon normalizes the momentum matrix by its Frobenius norm and then uses the scalar map  $\sigma \mapsto \sigma^p$  on the normalized singular values. The sweep  $p \in \{0.25, 0.5, 0.75\}$  moves from strong to mild homogenization: smaller  $p$  reduces singular-value ratios more aggressively while preserving the singular-vector directions (Fig. 5). The bowls in Fig. 6 back the monotone claim of §IV: final losses are 3.6520, 3.7119, and 3.8630 for  $p = 0.25, 0.5, 0.75$ , all selecting  $\eta = 0.05$  within the sampled grid. Even the best fixed-power arm found in our grid ( $p = 0.25$ ,  $\eta = 0.05$ , loss 3.6520) remains  $\sim 0.006$  above tuned Muon ( $3.6456 \pm 0.0008$ ), a gap roughly two to six times the run-to-run seed noise observed on identical configurations (0.001–0.003).

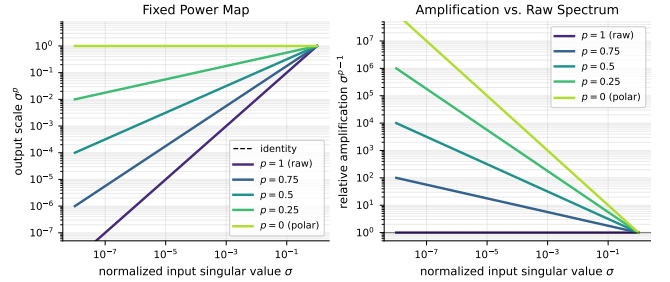


Fig. 5. Fixed-power singular-value maps used for HomogeneousMuon. Smaller exponents make the nonzero singular values closer to each other, while  $p = 0.75$  stays closer to normalized momentum.

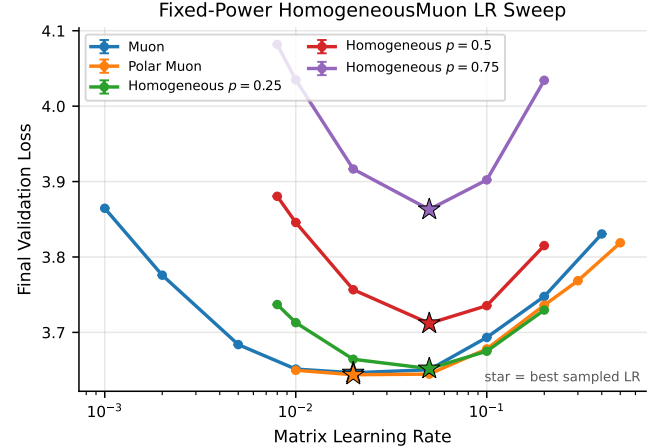


Fig. 6. Learning-rate bowls for Muon, exact-polar Muon, and fixed-power HomogeneousMuon. All three HomogeneousMuon exponents choose  $\eta = 0.05$  in the sampled grid, but stronger homogenization ( $p = 0.25$ ) is best among the fixed-power arms and approaches the flat-polar baselines most closely.

### E. Routing: where it helps and why

This subsection backs the routing claim of §IV: no improvement at a tuned rate, real protection at a mistuned one. The router computes a standardized layer statistic

$$z_{t,\ell} = \frac{x_{t,\ell} - \bar{x}_t}{\bar{\sigma}_t} \quad (8)$$



(where  $\bar{x}_t = \frac{1}{L} \sum_{\ell} x_{t,\ell}$  is the cross-layer mean of the proxy at step  $t$  and  $\tilde{\sigma}_t$  is its cross-layer standard deviation,  $\tilde{\sigma}_t = \sqrt{\frac{1}{L} \sum_{\ell} (x_{t,\ell} - \bar{x}_t)^2}$ , with a small floor added for numerical stability) and applies a bounded offset  $\Delta p_{t,\ell} = \text{clip}(\beta z_{t,\ell}, \pm \ell_{\max})$ ; the proxies themselves are defined and discussed in Appendix A-F. With a positive gain, layers whose proxy sits above the cross-layer average are pushed toward the raw-momentum side of the schedule, and layers below it toward flatter Muon-like updates, a regularized per-layer step-size rule.

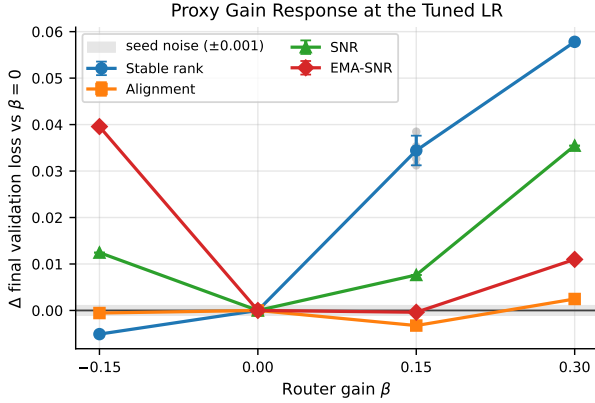


Fig. 7. Proxy gain response at the tuned LR (0.02). The axis is final validation loss relative to the no-route  $\beta = 0$  arm; the gray band marks the empirical seed-noise scale. Stable rank degrades monotonically with gain, alignment is the only proxy that stays near the baseline, and the two SNR signs are both worse. Lines connect the available tested gains, not dense sweeps for every proxy.

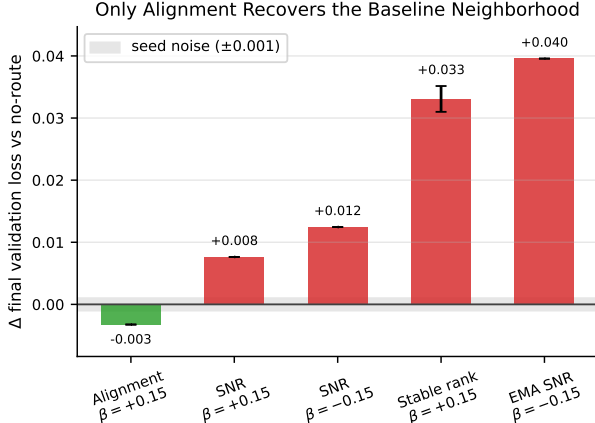


Fig. 8. Routing proxy comparison at the tuned LR. Values are validation-loss deltas from the no-route baseline, so lower is better. Alignment is the only proxy that reaches the baseline neighborhood; SNR, EMA-SNR, and stable rank are outside the seed-noise band on the worse side.

Fig. 9 shows why the proxy choice matters. The routed clock is not just a scalar gain shared across proxies: different statistics move different depths and matrix types. The useful alignment proxy has a milder pattern with clear MLP-output structure, while the worse SNR-family proxies concentrate

larger offsets in attention value/output matrices. Stable rank, in contrast, strongly raises late-block MLP-output exponents while lowering many value/output attention updates. This supports reading the proxy ablation as a signal-quality problem, not as a failure of the routing mechanism alone: the router can create structured layer-local changes, but most proxy signals choose unhelpful structures at the tuned LR.

Fig. 10 separates three effects that are otherwise easy to conflate. First, the router implementation itself is neutral: setting  $\beta = 0$  reproduces the DynMuon schedule within the observed run-to-run variation. Second, proxy choice matters more than router machinery. Alignment is selected as the only Route representative in the main table because it is the best completed proxy arm at the tuned LR, while stable rank and SNR-family arms are clearly worse. Third, the magnitude profile is not a nuisance term. The magnitude-decoupled grid keeps the spectral exponent changes but forces a flat polar Frobenius budget; its best completed point is 3.7345, clearly below the natural-magnitude route and DynMuon arms (boundary point of its grid, so a partially explored sweep).

This also explains why routing helped most at a deliberately high global LR. When the global LR is mistuned, layer-local conservative offsets can reduce damage in sensitive matrices. At the tuned LR, however, the same offsets can move the optimizer away from the already good global schedule. The structured offsets in Fig. 9 support the same reading: routing behaves as a layer-local step-size controller, but the proxy must choose the right layers and matrix types.

### F. Routing proxies

The router is a mechanism for converting a layer statistic into an exponent offset. The statistic is not assumed to be a curvature estimate; it is treated as a probe whose usefulness must be checked experimentally.

**Stable rank.** The cheapest proxy (as it is available from the Gram matrix the optimizer already forms) is

$$\text{sr}(M) = \frac{\|M\|_F^2}{\|M\|_2^2}. \quad (9)$$

It measures how spread out the momentum energy is across singular modes. A large stable rank means many modes carry energy; a small stable rank means the update is dominated by a few modes. With positive  $\beta$ , high stable-rank layers move toward raw momentum and low stable-rank layers move toward flatter Muon-like updates.

**Instantaneous SNR.** The direct SNR proxy is

$$\gamma_t = \frac{\|M_t\|_F}{\|G_t - M_t\|_F + \varepsilon}. \quad (10)$$

It compares the lookahead momentum direction with the current gradient. Larger values indicate that the current gradient and momentum are more consistent. The principled orientation is negative  $\beta$ : noisy layers with low SNR should be kept closer to raw momentum, while high-SNR layers can accept flatter updates. Empirically the orientation does not matter, as the completed arms give +0.008 ( $\beta = +0.15$ ) and +0.012

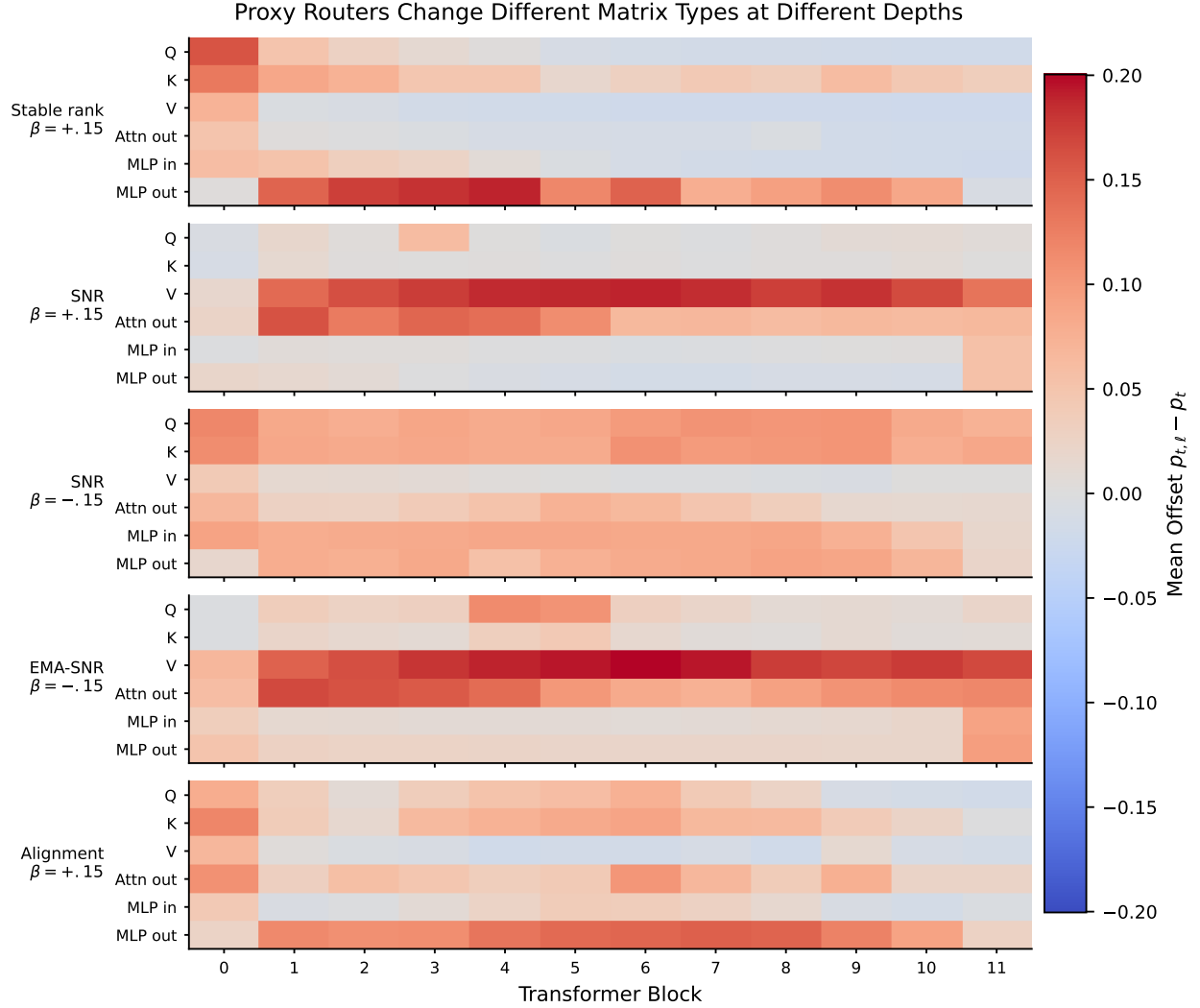


Fig. 9. Depth-by-matrix summary for all tuned-LR proxy routers. Each strip is one proxy/sign choice; each cell is the mean routed exponent offset  $p_{t,\ell} - p_t$  for one matrix type at one transformer block. This keeps the depth/block-type view without adding one full heatmap per proxy.

( $\beta = -0.15$ ) against the no-route baseline, both on the worse side (Fig. 8), which we read as evidence that this proxy does not act through its theorized mechanism in this setting.

**EMA SNR.** The EMA-SNR proxy uses a bias-corrected exponential moving average of the gradient and compares the estimated signal energy with the residual variance. It is meant to reduce the step-to-step noise of the direct SNR proxy. It should be judged against the direct SNR arm rather than against stable rank alone, because it measures temporal consistency rather than spectral spread.

**Alignment.** The alignment proxy is

$$\alpha_t = \frac{|\langle W_t, M_t \rangle|}{\|W_t\|_F \|M_t\|_F + \varepsilon}. \quad (11)$$

It measures whether the proposed update is aligned with the current weight in Frobenius inner product. A high value means the update points strongly along the existing parameter direction. This is not the same signal as RelMuon: alignment

is a scalar routing statistic, while RelMuon changes the whole singular-value vector using the weight spectrum.

### G. Spectrum controls

This subsection backs the magnitude-vs-shape claim of §IV. The spectrum controls test whether Muon’s flat singular values are themselves the critical ingredient. They keep the same momentum singular directions and roughly the same update budget, but change the values assigned to those directions. The useful contrast is not between Muon and a single alternative: it is between harmless changes that preserve the signal ordering and destructive changes that invert it.

All controls still have the form  $D = U \text{diag}(s) V^\top$  for the momentum SVD  $M = U \text{diag}(\sigma) V^\top$ . The flat-polar control uses  $s_i = 1$ . The random and inverted controls use the same polar-Frobenius budget as Muon: for any candidate vector  $a$ , define

$$\mathcal{N}(a) = \sqrt{k} a / \|a\|_2. \quad (12)$$



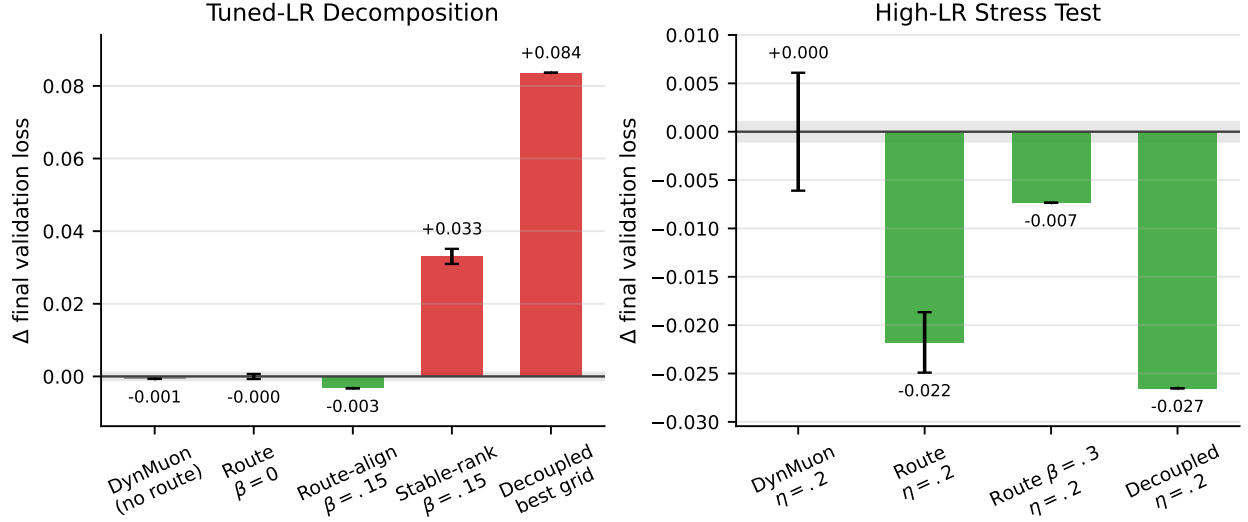


Fig. 10. Route decomposition ablation. Left: at the tuned LR, zero-gain routing matches DynMuon, alignment routing is the only useful proxy, stable-rank routing is worse, and removing the natural DynMuon magnitude profile is much worse. Right: at a deliberately high LR, routing and decoupling can reduce damage relative to high-LR DynMuon, which supports the interpretation of routing as a per-layer step-size rescue rather than a new tuned optimum.

Then random uses  $s = \mathcal{N}(r)$  with  $r_i \sim U(0, 1)$ , while inverted uses  $s = \mathcal{N}(\sigma^{\leftarrow})$ , where  $\sigma_i^{\leftarrow} = \sigma_{k+1-i}$ . Thus random tests whether the exact values matter, and inverted tests whether the largest scale must stay on the strongest momentum direction.

Kaon is the only control that does not choose  $s$  by a direct SVD formula. It normalizes  $M$  by  $\|M\|_F$  and iterates the scalar map

$$q(x) = 4.1x(1 - x^2)^2 \quad (13)$$

on the singular values for five steps, followed by the paper’s output normalization  $1/1.175$ . Equivalently, if  $x_i = \sigma_i(M)/\|M\|_F$ , its spectral scales are proportional to  $q^{(5)}(x_i)/1.175$ .

The random-spectrum control reaches 3.6492, within a few thousandths of Muon’s  $3.6456 \pm 0.0008$ , and the exact flat polar control reaches 3.6543. Kaon-style shaping is worse but not catastrophic at 3.6938. The inverted control is the important failure case: it reaches only 3.9670 because it anti-aligns scale with momentum strength. Thus the data do not say that every singular value should be one. They say that a good optimizer must preserve the effective update scale and avoid giving the largest steps to the least supported singular directions.

#### H. RelMuon and attention-only RelMuon

This subsection backs the weight-derived-spectrum claim of §IV. RelMuon keeps the singular directions of the momentum update but takes its scales from the current weight spectrum. In the log1p variant,

$$s_i = \frac{\log(1 + \sigma_i(W)) + \varepsilon}{\sqrt{k^{-1} \sum_j \log(1 + \sigma_j(W))^2 + \varepsilon}}. \quad (14)$$

The RMS normalization keeps the average squared scale near one, so RelMuon redistributes Muon’s update budget rather

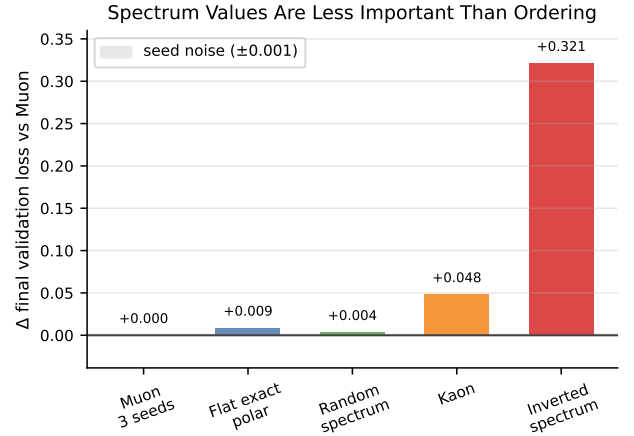


Fig. 11. Magnitude-matched spectrum controls at the tuned Muon LR. Randomizing spectrum values is close to Muon, while assigning the largest scales to the weakest momentum directions is destructive. This supports a weaker claim than “Muon’s flat spectrum is uniquely optimal”: effective magnitude and ordering matter more than the exact singular values.

than simply increasing it. The trust-ratio analogy is local to the spectrum: if weight and update singular directions were aligned, larger weight modes would receive proportionally larger update modes. In the implemented weight-only variant the pairing is by rank, not geometry, so the claim should remain modest.

The attention-only ablation applies this RelMuon update only to attention projection matrices and sends MLP matrices through AdamW. It is not meant to prove that attention is the only important component. It tests whether the expensive spectral update has most of its value in the token-mixing matrices of the transformer architecture. The correct comparison

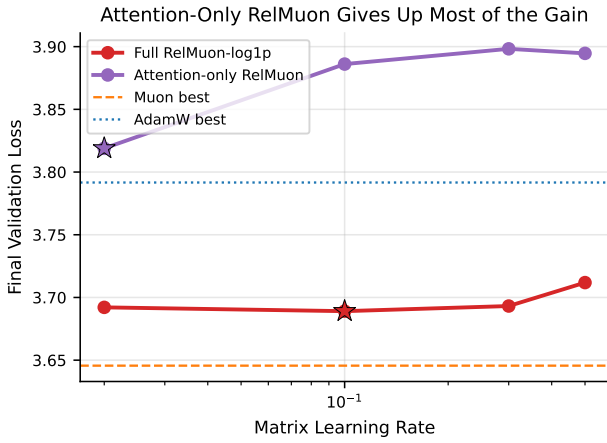


Fig. 12. Full RelMuon-log1p vs. attention-only RelMuon-log1p on the same completed LR grid, with tuned Muon and AdamW shown as references. Attention-only RelMuon loses most of full RelMuon’s gain while remaining much slower than AdamW.

is full RelMuon-log1p and attention-only RelMuon-log1p at the same LRs, same token budget, and same W&B project.

The completed comparison is negative for the efficiency hypothesis. Full RelMuon-log1p is best at  $\eta = 0.1$  with final validation loss 3.6891 in the spatial comparison sweep, consistent with the bowl result 3.6883. The attention-only variant is best at  $\eta = 0.02$  with 3.8189, close to AdamW’s 3.7917 and far from full RelMuon. It also costs 8.24s/step at its best LR, compared with 10.81s/step for full RelMuon-log1p and 2.29s/step for AdamW. The timing profile explains why: skipping the MLP matrices removes only part of the optimizer time, while forward/backward work and the attention SVDs remain. So attention-only RelMuon removes only part of the cost and removes most of the benefit.

### I. Cost comparison

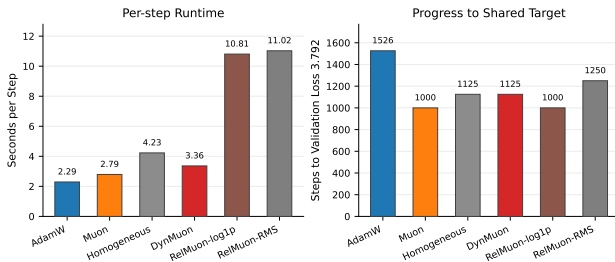


Fig. 13. Cost view of the selected representatives in Table III: seconds per optimizer step (left) and steps to a common validation target (right). Several spectra are acceptable in step count but not in wall-clock time; this figure backs the cost statements in §IV.

## APPENDIX B REPRODUCIBILITY

All reported numbers are tied to remote experiment logs, local history exports, and deterministic figure generation. The implementation includes validation checks for the algebraic

identities, optimizer-state restoration, and bitwise agreement with the released DynMuon baseline. The experiment launcher records the optimizer configuration, run group, random seed, learning rates, and train budget for each arm, so final tables can be regenerated from the logged runs rather than manually transcribed.